

Suwisa Aitha

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Summary

Electrical Engineering graduate student with hands-on experience in power system analysis, power quality simulation, and renewable energy integration. Experienced in supporting system reliability using MATLAB/Simulink, PowerWorld, and AutoCAD. Proven ability to translate analytical results into practical engineering insights to support system design and operational decision-making. FE/EIT in progress; open to relocation.

Education

George Mason University

Fairfax, VA

M.S. in Electrical Engineering

Jan 2025 – Expected May 2026

-Relevant Coursework: Power System Design, Power System Protection, Power Electronics

GPA: 3.07/4.00

Khon Kaen University

Khon Kaen, Thailand

B.S. in Electrical Engineering

Aug 2018 – Jul 2022

-Relevant Coursework: Electrical System Design, Electric Power Systems, Electrical Engineering Laboratory I-III

Experience

Electrical Design and Visualization

Bangkok, Thailand

B.S.E. Electronics Co., Ltd. — Cooperative Education Intern

Jul 2021 – Nov 2021

-Designed and tested ESP32 microcontroller circuits for real-time electrical data monitoring, supporting system validation and performance verification.

-Programmed Arduino and Unity to synchronize physical and virtual 3D system models, improving visualization efficiency by ~30% over 2D layouts.

-Created 3D electrical layout prototypes that improved coordination with design teams.

-Achieved ~87% alignment between physical and digital models, confirming system accuracy and reliability.

Selected Research Projects

Unified Power Quality Conditioner (UPQC) Simulation Project

George Mason University

-Developed a MATLAB/Simulink UPQC model to mitigate voltage sag and harmonic distortion, achieving approximately 50% reduction in THD, demonstrating effective power quality mitigation under disturbed operating conditions.

Power System Dynamics with Renewable Energy Penetration

George Mason University

-Modeled a single-machine power system in MATLAB/Simulink and analyzed 0%, 30%, and 60% renewable penetration via inertia reduction, showing deeper frequency nadirs, higher RoCoF, longer settling times, and larger rotor angle oscillations following a three-phase fault.

IoT-Based Air Quality Monitoring System

George Mason University

-Reviewed IoT-based embedded systems for PM2.5 and gas detection in Chiang Mai, Thailand, identifying peak PM2.5 levels of $187.6 \mu\text{g}/\text{m}^3$ (over $12\times$ WHO limits).

-Proposed a low-cost, scalable, and energy-efficient sensor concept that reduced estimated cost by $\approx 25\%$ and extended battery life to support community health and sustainability.

Certifications

-FE Exam in Progress — Target EIT Certification by May 2026

Skills

-Engineering Tools: MATLAB/Simulink, PowerWorld, AutoCAD, Revit

-Embedded / Visualization: Arduino, ESP32, Blender

-Programming: Python